



FACULTY OF COMPUTER SCIENCE AND ENGINEERING
Ghulam Ishaq Khan Institute of Engineering Sciences and Technology, Topi

Time: 15 mins.

CS437 Data Warehousing and Data Mining

Marks: 10

Instructor: Dr. Ayaz Umer

Dated: 07/05/2025

Name: _____

Reg#: _____

Quiz 4 (CLO-3-----Evaluate-----C5)

Question:

Marks [8+2]

Part A). Support vector machines learn a decision boundary that maximizes the margin between classes. You are training an SVM on a tiny dataset with 4 points shown in Figure 1, where two examples have class label -1 (denoted with plus) and two examples have class label +1 (denoted with triangles).

Evaluate the suitability of the resulting weight vector w and bias b obtained by applying SVM to this dataset. Justify whether the resulting decision boundary effectively separates the classes and maximizes the margin.

Part B). Clearly circle the support vectors and draw a decision boundary.

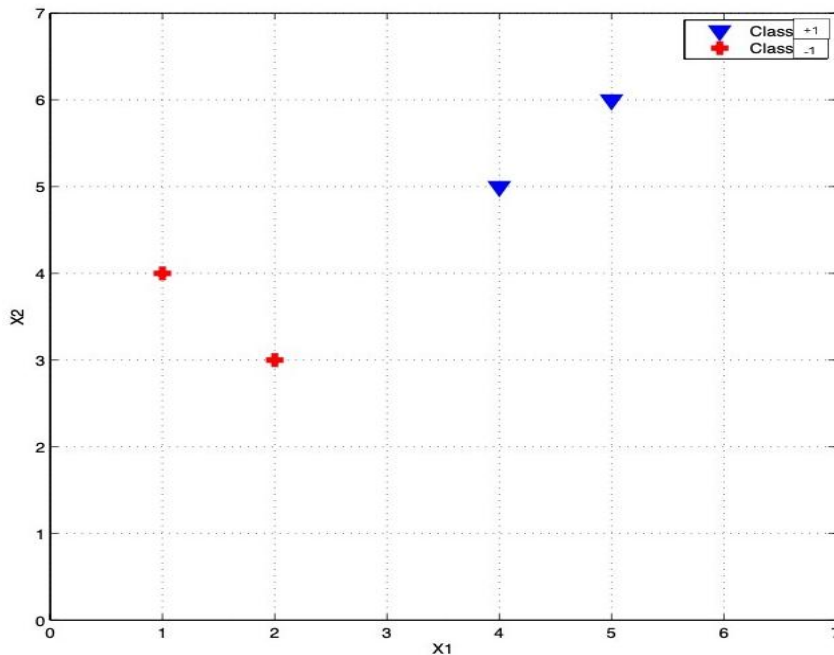


Figure 1

Answer: